

IMPLEMENTATION AND ANALYSIS OF 6-TRANSISTOR SRAM CELL

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Abstract:

Static Random-Access Memory (SRAM) stands as a vital component, offering swift and efficient on-chip data storage. It is made up of six transistors that are placed in a precise way to achieve stability and the capacity to read and write data. This 6-transistor SRAM cell plays a vital role, particularly in high-performance memory applications like cache memory. The main objective of this project is to focus on examining the performance, stability, and energy efficiency of the 6T SRAM cell. Extensive simulations are helpful to evaluate key performance parameters such as access time, write time, and power consumption. The primary aim is to identify any design limitations or flaws and propose potential enhancements. Factors like noise tolerance, resistance to data corruption during read operations, and ease of writing data should be analysed under different conditions. As power consumption is a critical concern in modern semiconductor devices, we assess the energy efficiency of the 6T

SRAM cell. This includes analyse the dynamic power under different operating scenarios. We will analyse its layouts and in addition to characterizing the 6T SRAM cell, we compare its performance and energy efficiency with other popular SRAM cell designs, such as 4T and 8T cells, to identify trade-offs and advantages.

KEYWORDS: 6T SRAM cell, 4T, 8T, Access time, Write time, Power consumption.

I. INTRODUCTION

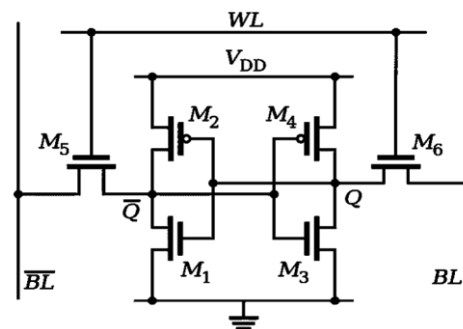


FIGURE 1: 6 TRANSISTOR STANDARD SRAM CELL.

The 6-transistor (6T) static random-access memory (SRAM) cell is a fundamental component in modern computing devices, serving as a tiny but crucial storage unit. Renowned for its reliability and speed, the 6T SRAM cell is utilized in a wide range of devices, from smartphones to powerful computers. Despite its small size, it offers exceptional performance, holding onto data for extended periods without the need for frequent refreshing. Comprising just six transistors, the 6T SRAM cell operates efficiently and effectively, ensuring quick data access and storage. Its compact design allows for seamless integration into various devices, contributing to their overall performance and responsiveness. Unlike other memory technologies, such as dynamic random-access memory (DRAM), the 6T SRAM cell does not require constant power to maintain stored data, making it an energy-efficient solution for memory storage.

The 6T SRAM cell functions by utilizing two cross-coupled inverters connected by two access transistors. These transistors form a latch-like structure that maintains the integrity of stored data. By carefully arranging the transistors and optimizing their sizes, designers ensure optimal performance while minimizing space and energy consumption. One key component of the 6T SRAM cell is the word line, which serves as a control signal for read and write operations. When the word line is activated, it enables the access transistors, allowing data to be read from or written into the cell. Conversely, when the word line is deactivated, the access transistors disconnect the cell from the bitlines, preserving the stored data. In terms of efficiency, the 6T SRAM cell excels due to its compact design and minimal power requirements. Designers carefully select the size and arrangement of each transistor to maximize performance while minimizing

energy consumption. This ensures that the cell operates reliably and efficiently, even in demanding applications. Overall, the 6T SRAM cell plays a vital role in modern memory systems, offering a combination of reliability, speed, and efficiency. Its compact design and efficient operation make it an ideal choice for various applications, from consumer electronics to high-performance computing. As technology continues to advance, the 6T SRAM cell is expected to remain a cornerstone of memory storage, providing fast and reliable data access for years to come.

II. LITERATURE REVIEW

Ezeogu, A., 2019. Performance Analysis of 6T and 9T SRAM. arXiv preprint arXiv:1905.08624[1]. This paper compares the performance of 6T and 9T SRAM memory cells in 45nm CMOS technology, evaluating stability, leakage, and process variations. It introduces the N-curve metric for stability measurement and demonstrates that the 9T SRAM design exhibits higher stability, lower leakage, and better immunity to process variations compared to the 6T SRAM. The article includes simulation results for static noise margin, write ability, and read stability, along with references to related research.

Rath, S. and Panda, S., 2017. Analysis of 6T SRAM cell in different technologies. Circulation in Computer Science, pp.7-10[2]. The paper presents a comprehensive analysis of a 6T SRAM cell in 180nm, 90nm, and 45nm technology nodes using Cadence Virtuoso tool. It includes the design, implementation, and performance evaluation of the SRAM cell in terms of speed, power dissipation, and static noise margin. The analysis demonstrates speed improvements, decreased delay time, and reduced power dissipation with scaling of

technology. Additionally, the paper discusses related research in the field, providing a valuable contribution to the understanding of SRAM cell performance across different technology nodes.

Shivaprakash, G. and Suresh, D.S., 2016. Design of low power 6T-SRAM cell and analysis for high-speed application. *Indian Journal of Science and Technology*, 9(46), pp.1-10[3]. The paper discusses the design and analysis of a low power 6T-SRAM cell for high-speed applications using 90nm technology. It covers topics such as Static Noise Margin, Data Retention Voltage, Read Margin, and Write Margin, and includes graphs and tables showing the relationship between various parameters. The study concludes that better Static Noise Margin can be achieved by sacrificing the area of the transistors and provides insights into the performance of the 6T-SRAM cell. Additionally, the article includes references to related research in the field.

Nigam, A.K., Singh, S. and Tiwari, A., 2015. 6T SRAM Cell: Design and Analysis. *Intl J Engg Sci Adv Research*, 1(2), pp.27-29[4]. The paper presents a comprehensive analysis of a 6T SRAM cell's performance in 45nm and 180nm technologies, focusing on power consumption, delay, and SNM. It includes a detailed explanation of the cell's operation and provides simulation results for both technologies. The conclusion emphasizes the trade-offs between different performance parameters and suggests future research directions for new SRAM cell designs. The paper also includes a list of references for additional reading.

Lakshmi, T.V. and Kamaraju, M., 2021, February. Implementation of high performance 6T-SRAM cell. In *Journal of Physics: Conference Series* (Vol. 1804, No. 1, p. 012185) [5]. IOP Publishing. Power reduction techniques in cell technology, low power VLSI circuit design, SRAM

analysis, and CMOS integrated circuit design are covered in this study. It compares power dissipation and delay performance using CMOS, Gated VDD, and MTCMOS techniques and focuses on the design of a high-performance 6T-SRAM cell using low power VLSI techniques. The results show that while Gated VDD also considerably lower power consumption, MTCMOS offers high speed while decreasing power consumption. The study comes to the conclusion that these methods can maximize SRAM cell performance.

Singh, S., Arora, N., Suthar, M. and Gupta, N., 2012. Performance evaluation of different SRAM cell structures at different technologies [6]. *International Journal of VLSI Design and Communication Systems*, 3(1), p.97. The paper explores the demand for low-power devices in VLSI circuits and the use of static random-access memory (SRAM). It compares the performance of various SRAM cell structures at different technology nodes, finding that the 10T Modified SRAM cell at 90nm and 45nm, and the 10T SRAM cell at 32nm, demonstrate superior power consumption and speed. The study utilized simulation on the Tanner EDA tool to reach these conclusions.

Balanarasimhulu, P. and Reddy, G.K., 2014. A Comparative Analysis of 6 Transistor SRAM and 6T ROM Embedded SRAM [7]. The paper presents a comparative analysis of different SRAM cell designs, including 6T SRAM, 6T RSRAM, and 8T RSRAM, with a focus on power comparisons. It introduces modified ROM embedded SRAM cells with the aim of reducing power consumption and area, and also proposes an 8T ROM Embedded SRAM design to address write stability issues. The simulations are conducted using LT SPICE Software, and the paper includes references to related research in the field.

Adisheshaiah, M., Rao, D.S.B. and Reddy, V.V.T., 2012. Implementation and desing of 6T-SRAM with read and write assists circuits [8]. *Int. J. Res. Eng. Appl. Sci.*, 2, pp.36-50. The implementation of read and write assist circuits for SRAM cells in 65nm technology is shown in the study along with a comparison to 120nm technology. Power dissipation, circuit layouts, schematics, and simulation results for 32-bit SRAM are covered. Topics such as power savings and difficulties in VLSI circuit design are discussed, as well as how well the aid circuitry controls power dissipation. The NOR decoder, the SRAM cell's behavior when a read fails, and the architecture and parts of the SRAM chip are all explained in detail. References and the project's future scope are included in the article's conclusion.

Khan, Q.M., Perdriau, R., Ramdani, M. and Koohestani, M., 2022. A comparative performance analysis of 6T and 9T SRAM integrated circuits: SOI vs. bulk [9]. *IEEE Letters on Electromagnetic Compatibility Practice and Applications*, 4(2), pp.25-30. The paper presents a comparative performance analysis of 6T and 9T static random-access memory (SRAM) cells in silicon-on-insulator (SOI) and bulk CMOS technologies. It evaluates data stability, power metrics, and resilience to process and temperature variations. The results indicate that SRAM cells designed with SOI exhibit lower static power dissipation and leakage current, higher data stability, lower write ability, larger sensitivity to process variations, and higher resilience to temperature deviations compared to bulk technology. The study provides valuable insights for the design and implementation of SRAM cells in different technologies.

III. METHODOLOGY

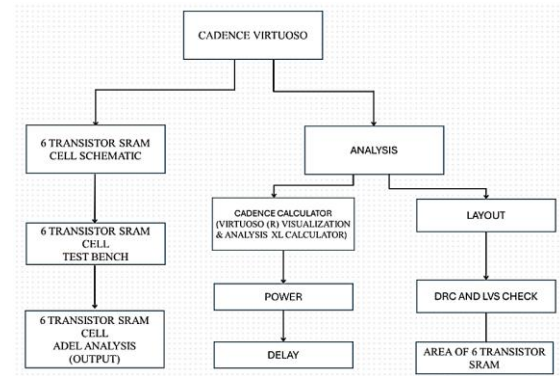


FIGURE 3:FLOW DIAGRAM OF CADENCE TOOL.

The design flow for the 6-transistor SRAM cell begins with creating a library linked to the existing gpdk45 library. In the cell view, we create the schematic of the 6T SRAM, including ports such as BL, BLB, WL, Q, QB, VDD, and GND. This schematic

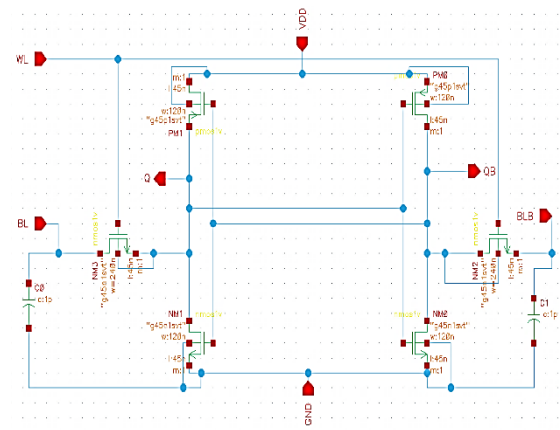


FIGURE 2: SCHEMATIC OF 6 TRANSISTOR SRAM.

comprises six transistors, with two serving as access transistors and four forming the inner memory unit known as cross-coupled inverters, which stores the data. After completing the schematic, we generate the symbol for the SRAM. In another cell view, we implement the test bench, leading to pre-layout simulations using ADEL window analysis. Then, using Cadence Virtuoso (r) Visualization & Analysis XL calculator, we analyze the power and delay using special functions in the XL calculator for the proposed SRAM. Once pre-layout simulations are complete, we move on to

designing the layout in Layout XL and perform area calculations for the SRAM layout. Finally, we conduct DRC and LVS checks.

IV. DESIGN ANALYSIS

The design analysis for a 6-transistor (6T) SRAM cell design are crucial for its successful implementation and performance optimization within memory systems. These specifications encompass several key aspects that define the characteristics and behaviour of the SRAM cell. Firstly, the choice of transistor type (pmos1v and nmos1v) and size (2:1 ratio), typically CMOS technology, along with specific dimensions, influences the cell's footprint on the integrated circuit (IC) chip and its performance. The supply voltage levels must be defined to ensure reliable operation, both under nominal conditions and at maximum allowable levels. In 45 nm technology we use 1.1V supply voltage. Access times for reading and writing data, along with retention time for maintaining stored data integrity, are critical parameters influencing the cell's performance and reliability. By adhering to these hardware specifications, designers can ensure the successful integration and optimization of 6T SRAM cell designs within memory systems, meeting performance, power, and reliability requirements.

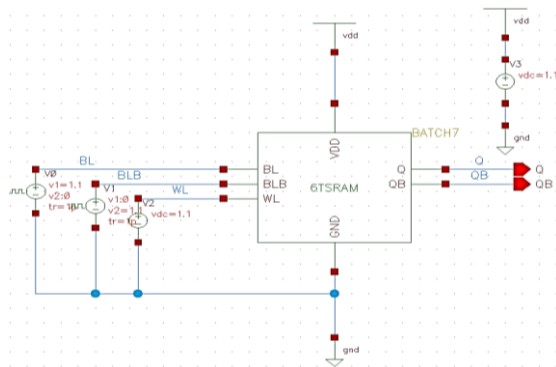


FIGURE 4: TEST BENCH FOR 6 TRANSISTOR SRAM SCHEMATIC.

1) READ OPERATION

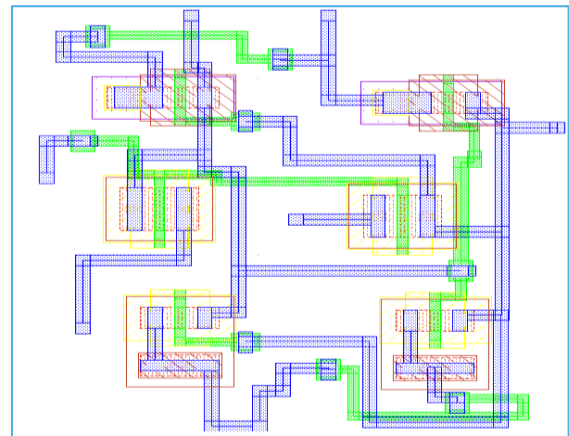


FIGURE 5: LAYOUT FOR 6 TRANSISTOR SRAM CELL.

During a read operation in a 6-transistor SRAM cell, the wordline (WL) is activated, enabling access transistors to connect the bitlines (BL and BLB) to the cross-coupled inverters. The data stored in the SRAM cell is then amplified and presented at the output nodes (Q and QB). These output nodes reflect the stored logic values, allowing external circuitry to read the data stored in the SRAM cell.

2) WRITE OPERATION

During a write operation in a 6-transistor SRAM cell, the desired data value is applied to the bitlines (BL and BLB). The wordline (WL) is activated to enable access transistors to connect the bitlines to the cross-coupled inverters. The data is then latched into the SRAM cell by the cross-coupled inverters, updating the stored value. Finally, the wordline is deactivated, preserving the written data until the next operation.

3) HOLD OPERATION

During a hold operation in a 6-transistor SRAM cell, the word line (WL) is deactivated, isolating the access transistors from the cross-coupled inverters. This prevents any changes to the stored data, maintaining the stability of the SRAM cell.

The data stored in the SRAM cell remains unaffected, allowing it to retain its logic value until the next read or write operation is initiated.

4) DELAY

The time interval that passes between applying an input and its return is known as the delay. Any system design's primary goal is to increase system speed. SRAM speed is measured by its read and write access times.

V. RESULT ANALYSIS

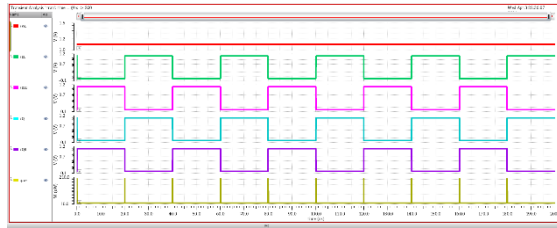


FIGURE 4: SIMULATION RESULTS OF 6T SRAM.

When the word line (WL) is set high (WL=1), it activates the access transistors within the SRAM cell. When the Bitline (BL) is high (BL=1), the Complementary Bitline (BLB) should be low (BLB=0). Conversely, when BL is low (BL=0), BLB should be high (BLB=1). This allows the bitlines (BL and BLB) to connect to the cross-coupled inverters. The behavior of the inner inverters, acting as the memory unit, depends on the voltages applied to BL and BLB. During a read operation, if the stored data (Q) is low ('0'), then its complement (QBAR) should be high ('1'). Conversely, if the stored data (Q) is high ('1'), then its complement (QBAR) should be low ('0').

1) LAYOUT DESIGN

Layout design involves translating the schematic of the 6-transistor SRAM cell into a physical representation, adhering to design rules. It focuses on optimizing transistor sizes and placement to

minimize area occupancy while ensuring proper routing and connectivity. DRC (Design Rule Check) and LVS (Layout vs. Schematic) checks are performed to ensure the layout's compliance with fabrication requirements and schematic integrity.

TABLE 1: FINAL RESULTS OF 6 TRANSISTOR SRAM CELL IN 45 NANO METER TECHNOLOGY.

SNO	PARAMETER	VALUE		TECHNOLOGY
1	POWER (nano watts)	28.05		45nm
2	DELAY (nano seconds)	BL vs Q	1.442	
		BL vs QB	7.073	
		BLB vs Q	1.351	
		BLB vs QB	6.983	
3	AREA (square nanometres)	11.155		

VI. CONCLUSION

The analysis of the 6-transistor SRAM using Cadence software in the 45-nanometer technology generation provided valuable insights into its performance. Power consumption varied between 28.05n watts and 74.169n watts across different simulations, with the lowest consumption observed at 28.05n watts, indicating improved efficiency under certain settings. Delay characterization revealed timing behavior ranging from 1.442 nanoseconds to 7.073 nanoseconds for different signal pairs, offering insights into data access speed. Layout design resulted in an area requirement of 11.155 square nanometers. This emphasizes the importance of optimizing design parameters for a balanced performance, power efficiency, and layout area in 45-nanometer technology for 6-transistor SRAM designs.

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